

TRI-SUL 480 PREMIX

(FOR VETERINARY USE ONLY)

COMPOSITION

Sulfamethazine	400 g
Trimethoprim	80 g
Exp qs	1000 g

PRODUCT DESCRIPTION

TRI-SUL 480 is a white-coloured powder.

PHARMACODYNAMIC

Alone, sulfonamides are bacteriostatic agents and trimethoprim is bactericidal, but in combination, the potentiated sulfas are bactericidal. Potentiated sulfas sequentially inhibit enzymes in the folic acid pathway, thereby inhibiting bacterial thymidine synthesis. The sulfonamide blocks the conversion of para-aminobenzoic acid (PABA) to dihydrofolic acid (DFA), and trimethoprim blocks the conversion of DFA to tetrahydrofolic acid by inhibiting dihydrofolate reductase. The in vitro optimal ratio for most susceptible bacteria is approximately 1:20 (trimethoprim:sulfa), but synergistic activity can reportedly occur with ratios of 1:1 - 1:40. The serum concentration of the trimethoprim component is considered to be more important than the sulfa concentration. For most susceptible bacteria, the MIC's for TMP are generally above 0.5 micrograms/ml. The potentiated sulfas have a fairly broad spectrum of activity. Gram positive bacteria that are generally susceptible include most streptococci, many strains of staphylococcus and Nocardia. Approximately 30% of strains tested of Streptococcus zooepidemicus are resistant to TMP/Sulfa in horses. Many gram negative organisms of the family Enterobacteriaceae are susceptible to the potentiated sulfas, but not Pseudomonas aeruginosa. Some protozoa (Pneumocystis carinii, Coccidia and Toxoplasma) are also inhibited by the combination. Potentiated sulfas reportedly have little activity against most anaerobes, but opinions on this

varies. Resistance will develop slower to the combination of drugs than to either one alone. In gram negative organisms, resistance is usually plasmid-mediated.

PHARMACOKINETIC

Trimethoprim/sulfa is well absorbed after oral administration, with peak levels occurring about 1-4 hours after dosing. The drug is more slowly absorbed after subcutaneous absorption, however. In ruminants, the trimethoprim is apparently trapped in the rumenoreticulum after oral administration and undergoes some degradation.

Trimethoprim/sulfa is well distributed in the body. When meninges are inflamed, the drugs enter the CSF in levels of about 50% of those found in the serum. Both drugs cross the placenta and are distributed into milk. Trimethoprim/sulfa is both renally excreted unchanged via glomerular filtration and tubular secretion and metabolized by the liver. The sulfas are primarily acetylated and conjugated with glucuronic acid and trimethoprim is metabolized to oxide and hydroxylated metabolites.

Trimethoprim may be more extensively metabolized by the liver in adult ruminants than in other species. The serum elimination half-lives for trimethoprim in various species are: 1.5 hours (cattle). The serum elimination half-lives for sulfadiazine in various species are: 2.5 hours (cattle). While trimethoprim is quite rapidly eliminated from the serum, the drug may persist for a longer period of time in tissues.

INDICATIONS

For the treatment of respiratory, enteric or urinary tract infections in calves, pigs and poultry due to organisms susceptible to the combination of sulfadimidine and trimethoprim, or where such organisms are implicated as secondary pathogens.

CONTRA-INDICATIONS

Do not administer to animals with known sulfonamide sensitivity, liver parenchymal damage or blood dyscrasia.



SHENNONG ANIMAL HEALTH (M) SDN BHD (441647-X)

神農動物保健(馬)有限公司

No. 2, 4, 6, 8 & 10 Jln Industri USJ 1/19, Tmn Perindustrian USJ 1, 47610 Subang Jaya, Selangor, Malaysia.

ADMINISTRATION & DOSAGE

Calves: Use 3 g this product per 50 kg bodyweight daily. Administer on dampened food or mixed with honey or molasses.

Pigs: Use 3 g this product per 50 kg bodyweight daily.

Poultry: Use 3 to 6 g this product per 12 L drinking water for 3 to 5 days.

Administer mixed thoroughly with food or in drinking water.

Any variation by the prescribing veterinarian to the approved dose, frequency, duration, route, and disease or target species may require extending the approved withholding period.

INTERACTION WITH OTHER MEDICAMENTS

In the absence of compatibility studies, this veterinary medicinal product must not be mixed with other products.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

Pregnancy/Reproduction:

Sulfonamides cross the placenta in pregnant animals.

Some teratogenic effects have been seen when very high doses were given to pregnant mice and rats.

Lactation:

Sulfonamides are distributed into milk; however, the sulfonamides that are clinically relevant to food-producing animals are distributed into milk in concentrations too low to be therapeutic but high enough to produce residues. Sulfadiazine and sulfanilamide are more efficiently distributed into milk than most sulfonamides, but are not used in dairy cattle. For many sulfonamides, 0.5 to 2% of the total dose is found in the milk. Distribution into milk varies depending on the amount of non-protein-bound sulfonamide present in the blood and the amount of the nonionized and therefore liposoluble form of the medication present. Sulfonamides with higher pKa values produce a higher proportion of drug in the blood that is nonionized, and if

other factors, such as the rate of biotransformation, also support it, may be distributed more easily into milk. For lactating dairy cattle, concentration of the active parent compound of sulfamethazine, measured at a specific time in milk, is about 20% of the concentration in the blood.

OVERDOSE AND TREATMENT

Stop medication and to seek for veterinarian advice if there is sign of over dosage.

SIDE EFFECT

Nausea, upset stomach, loss of appetite.

WITHDRAWAL TIME

Meat (CALVES, PIGS, POULTRY): Do not use less than 14 days before slaughter for human consumption. Eggs: Do not use in laying birds where eggs or egg products are to be used for human consumption or processing.

STORAGE

Store in cool dry place (below 25°C). Strictly avoid light and heat exposure.

SHELF LIFE

24 months from manufacture date.

SHELF LIFE after first opening

12 months. Keep in cool dry place (below 25°C) and airtight.

WARNING AND PRECAUTIONS

- For Veterinary use only.
- Keep out of reach of children.
- Jauhkan ubat dari kanak-kanak.

PACKAGING

The powder filled 15 kg or 20 kg in white PE plastic drum, by automatic filling machine.

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